Task 4: Classification

Validated Encoders

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1 Dataset Preparation

1.1 Label Aggregation

The annotation tensor has shape $[T, C, A]$, with each entry encoding the temporal overlap $\in [0, 1]$ between an annotated event and a one-second segment. We convert it to binary labels $[T, C]$ in two steps. First, we binarize per annotator at τ=0.5: annotator a marks class c present in segment t when the overlap exceeds 0.5. Second, we combine annotators by strict majority vote: the label is 1 only if more than $A/2$ annotators agree, with ties resolving to 0. Across all 3,656 recordings (168,239 segments) this gives 53.7 % single-class, 11.6 % multi-class, and 34.7 % silent segments.

This strategy is robust to one noisy annotator and biases labels toward precision, which is appropriate when label noise hurts a classifier more than missed positives. Three known limitations apply: the number of annotators varies (A=1 in 17.1 % of files, A=2 in 47.9 %, $A\geq$3 in 35 %), so single-annotator files get no aggregation and two-annotator files degenerate to unanimous consent; the τ=0.5 cut disfavors very brief events (`light_switch` ends up at $0.30\,\%$ positive rate); and soft information is discarded, since overlap pairs $(0.9, 0.45)$ and $(0.51, 0.51)$ yield identical labels.

1.2 Data Split

We could identify two leakage sources; *Segment level*: adjacent one-second windows from the same recording share the same device, environment, and (at the 0.5 s hop) acoustic content. *Collector level*: each contributer/ data collector contributed up to ten recordings sharing microphone, placement, and room, so segments from the same collector stay correlated across different files. Splitting by recording alone would not block the second mechanism, so we split by `collector_id` with `GroupShuffleSplit`: 20 % of collectors form test, 25 % of the remainder form validation (60/20/20 overall). The empirical sizes are 244 / 82 / 82 collectors, or 100,166 / 34,245 / 33,828 segments; zero collector overlap was verified.

Similar class distributions across splits matter so that val-tuned hyperparameters and thresholds transfer to test. Multi-label stratification under a group constraint has no scikit-learn primitive, so we sweep 20 random seeds and pick the one that minimizes the symmetric KL divergence between per class positive rates of train vs. val and train vs. test. The chosen split -seed 9- reaches a total symmetric KL of 0.0054; per-class rates deviate by at most 1.5 pp (Table 2). The max to min positive ratio on train is roughly $47\times$ and the dominant residual imbalance is `light_switch`, with only 88 test positives.

1.3 Preprocessing

Two transforms are fit on train only. `log1p` on the `energy` and `power` columns (8 of 960 dimensions) handles their heavy right tails: `power_max` reached 6.79×10^4 versus a mean near 112, giving post-standardization outliers at 80σ; after `log1p` those columns peak at 11.13. We then z-score standardize; per-column $|z|_{\max}$ drops from 80.07 to 21.77 (median 6.90). Scaling is applied per classifier, as tree models are scale-invariant. For CNN/CRNN we cache 128×100 log-mel spectrograms; for AST we cache 16 kHz waveforms and use HuggingFace's AudioSet normalization. We did NOT apply explicit feature selection: the 960-D vector contains correlated blocks (`melspect`, `mfcc`), but per classifier behavior differs, so selection was tested downstream during the RF study (Section 3.3).

2 Evaluation

2.1 Evaluation Criterion

We use **macro F1** as the primary metric, with micro F1 and weighted F1 as diagnostics. Task 3 showed the dataset is highly imbalanced ($47\times$ gap between most and least frequent classes), and under such imbalance accuracy and micro F1 are dominated by common classes, so a model that ignores rare classes can still score well. Macro F1 weighs all 15 classes equally and is comparable across model families because it depends only on per class confusions.

2.2 Baseline and Achievable Performance

The floor baseline is an *empirical prior* that predicts class c with its training positive rate ignoring the input features. On test, it reaches macro F1 = **0.053**. As a sanity check, scikit-learn's `DummyClassifier` under stratified, most-frequent, uniform, and constant strategies stays at or below 0.10; the most-frequent variant reaches 0.35 *subset accuracy* but 0.00 macro F1, confirming why accuracy is the wrong metric here. Any trained model that does not outperform the reached 0.053 is useless.

Perfect macro F1 is not achievable, bounded by three ceilings. The *label-noise* ceiling: 17.1 % of files are unfiltered (single annotator) and the test ground truth itself contains noise. The *rare-class* ceiling: `light_switch` has only 88 test positives, so a handful of misses swings macro F1 sharply. The *representation* ceiling: the 960-D vector aggregates 32 ms frames over one-second windows using mean/std/min/max, which low-pass filters temporal structure and erases the brief transients that distinguish events like `light_switch` or `door_open_close`. Our results below suggest an achievable range of approximately 0.55–0.61 macro-F1 on this split.

3 Experiments

3.1 Protocol and Class Imbalance Handling

Every model is fit on train, selected on validation, and evaluated once on test at the very end. Tree based and linear models select by val macro F1; for neural net loss variants we select by val mean average precision (threshold-free), tune per-class thresholds on val, and only then evaluate on test. Imbalance is addressed on the loss and decision sides, not by feature-domain resampling: `class_weight='balanced'` for sklearn models; `pos_weight` = $\mathrm{sqrt}(1/\mathrm{freq})$ clipped to $[0.5, 30]$ for BCE losses; focal loss (Lin et al., 2017) with γ=2 for one neural config; and per-class threshold tuning on val (Table 1).

3.2 Model Suite

Seven classifiers across three input modalities. *On the 960-D vector*: Logistic Regression (one-vs-rest, `saga`, L2), Random Forest (200 trees, no depth limit, leaf 2), XGBoost (300 trees, depth 6, `hist`), and MLP-v2 (960→512→256→128→15, BN, dropout 0.3, ∼691 k params). *On* 128×100 *log-mel*: CNN (four conv blocks + GAP, ∼392 k), CRNN (two conv blocks + two-layer BiLSTM, ∼1.65 M). *On raw 16 kHz audio*: AST (Gong et al., 2021) as a linear probe with frozen AudioSet (Gemmeke et al., 2017) backbone and an 11 k-param `Linear(768, 15)` head trained on mean-pooled embeddings (full fine tuning was infeasible on our Apple MPS machine). LogReg and RF are our two chosen "different model classes" for question 3(a); the rest extend the comparison.

3.3 Hyperparameter Studies

Threshold tuning is the dominant lever. For every probabilistic model, the per-class decision threshold matters more than any architectural hyperparameter. Table 1 shows that moving from a fixed 0.5 cut-off to per-class val-tuned thresholds increases val macro F1 by $+0.18$ for LogReg and $+0.21$ for RF; the lift is smaller for XGBoost ($+0.08$) and MLP-v2 ($\sim +0.01$), possibly because they produce better calibrated probabilities. Selected thresholds are not comparable across families: LogReg lands near $[0.75, 0.95]$, RF near $[0.05, 0.37]$.

Logistic Regression. Relevant hyperparameters are regularization strength C, penalty type, and the class-weighting scheme. With L2 and `saga`, val macro-F1 is stable over $C \in \{0.1, 1, 10\}$ (variation under 0.005), so we keep C=1. Removing `class_weight='balanced'` hurts rare-class F1; switching to L1 gives no gain at higher cost. The decisive lift comes from per-class threshold tuning ($+0.18$). Test macro F1 is 0.4723, clearly above the 0.053 floor.

Table 1: Per-class threshold tuning is the largest decision-rule lever; values are validation macro-F1.

Model	**val @ 0.5**	**val (tuned)**	Δ
Logistic Regression	0.327	0.510	$+0.183$
Random Forest	0.292	0.497	$+0.205$
XGBoost	0.471	0.552	$+0.081$
MLP v2	$\sim$0.55	$\sim$0.56	$+0.01$

Random Forest. Relevant hyperparameters are tree depth, leaf size, the per-split feature subsample, and class weighting. We ran a 28-configuration random search under `GroupKFold` (5 folds on `collector_id`) over `max_depth` $\in \{10, 20, 30, 50, \text{None}\}$, `min_samples_leaf` $\in \{1, 2, 4, 8, 16\}$, `max_features` $\in \{\text{sqrt}, \text{log2}, 0.2, 0.4, 0.6\}$, and `class_weight` $\in \{\text{balanced}, \text{balanced_subsample}, \text{None}\}$. Ranking was performed on grouped 8k/12k subsamples (stable under subsampling); the winner was refit on the full 100k partition. Grouped-CV macro F1 moved from 0.452 (baseline config) to 0.466 (winner), with the top three configurations spanning $[0.459, 0.466]$. On the same model, per-class threshold tuning on out-of-fold predictions moved CV macro-F1 from 0.223 to 0.466, a $+0.243$ lever about $17\times$ larger than the hyperparameter search. Correlation filtering at $|r| > 0.95$ was neutral; PCA hurt by -0.06 ; importance pruning to 480 features gave $+0.014$. ExtraTrees (0.438) and one-vs-rest BalancedRandomForest (0.429) were slightly worse. The refit final RF reaches test macro F1 = **0.5116**, statistically tied with the baseline 0.5040.

Other models: XGBoost (300 trees, depth 6, lr 0.1) reaches 0.5605. MLP-v2 trained under three losses (BCE, pos-weighted BCE, focal γ=2); pos-weighted BCE was selected by val mean-AP and reaches 0.5495. CNN and CRNN were tuned for learning rate and dropout, both reaching $\sim$0.60. AST was used as a linear probe with no further tuning.

3.4 Final Comparison

Table 3 shows the test leader board, with the per-class F1 breakdown of every model in Figure 1. The progression from prior to linear to trees to MLP to pretrained linear probe to from-scratch deep nets on spectrograms is monotonic and consistent with theory. CRNN and CNN are statistically tied (gap 0.0013, within the ± 0.005 seed-variance floor). Aggregated-feature models plateau near 0.56 (XGBoost), the representation ceiling; CNN and CRNN clear it by $+0.045$ because raw spectrograms preserve the temporal structure aggregation removes. The val-to-test gap is at most 0.01 for every strong model, sometimes negative (XGBoost val 0.552, test 0.560), so the collector-grouped split generalizes stably.

Both required classifiers beat the baseline by a large margin (LogReg by $+0.42$, RF by $+0.45$ over the 0.053 prior). LogReg is cheap and interpretable but capacity-limited and reliant on threshold tuning; RF captures non linear feature interactions and is scale robust but needs aggressive threshold calibration. The RF-to-CNN gap shows that the input representation, not the classifier, is the real bottleneck. AST beats every feature-engineered model despite a frozen backbone (AudioSet covers most of our 15 classes) but loses to CNN/CRNN because of the YouTube vs. close-mic domain gap and the linear-only head.

3.5 Strategy B: Augmentation Null Result

We also trained the four aggregated-feature models on a parallel set augmented with pitch shift, time stretch, and noise (24,224 class-positive segments from 5,205 augmented files, concentrated on the rarest classes). All test deltas (Table 3) lie in $[-0.006, +0.004]$, within seed variance. The mechanism is mechanistic: mean/std/min/max over a one-second window is a low-pass filter on time-domain perturbations, so an augmented vector behaves like a duplicated sample with mild jitter, the same effect `class_weight='balanced'` already produces.

4 Case Study and Reflection

4.1 Case Study

We examine the headline CNN on two test recordings (Figures 2 and 3). Files are ranked by macro-F1 over the active or predicted classes, so the score reflects this file rather than averaging over inactive classes.

Successful example: `000727.wav` (47 segments, 3 active classes, active-class F1 = **0.930**). The three sustained classes `cutlery_dishes` (F1 = 0.82), `microwave` (0.99), and `running_water` (0.98) are all detected, with per-segment probability that hug the ground-truth boxes. The other seven shown classes stay below threshold.

Failure example: `005084.wav` (34 segments, 4 active classes, active-class F1 = **0.000**). The single true class is `window_open_close` (ground truth $\sim$8–11 s), and its probability never lifts. The model instead hallucinates `door_open_close` (15 FP), `footsteps` (6 FP), and `wardrobe_drawer_open_close` (1 FP), confusing the window-event impulses with other brief mechanical transients. This is the failure mode predicted by Figure 1.

4.2 Reflection

We can conclude that the aggregated feature representation, not the classifier, is the dominant bottleneck. Aggregated feature models plateau around 0.55 even after fitting, tuning, threshold correction, and augmentation; spectrogram models clear that ceiling by $\sim$0.05; the AST linear probe sits between the two. The per-class pattern (Figure 1) is the signature of this ceiling, not a defect of any single model: sustained broadband sounds survive one-second aggregation and are detected reliably by every model, while brief transients lose their timing cues and become acoustically indistinguishable, so `window_open_close` confuses with `door_open_close` and `wardrobe_drawer_open_close`. The three ceilings identified in Section 2.2 (label noise, rare classes, representation) still apply; a full AST fine-tune would likely beat CRNN's 0.605 but was infeasible on our Apple MPS. On the held out challenge set we expect the headline model in the 0.55–0.61 range; the val-to-test gap is near zero, so threshold transfer is the main remaining risk.

5 Figures and Tables

Table 2: Per-class positive rate (%) across the train/val/test splits. Deviations are at most 1.5 pp; total symmetric KL is 0.0054.

Class	Train	Val	Test
running_water	12.82	12.12	12.39
footsteps	11.41	11.78	12.58
keyboard_typing	9.11	9.07	9.85
microwave	6.81	6.58	7.62
vacuum_cleaner	6.39	7.13	5.89
phone_ringing	6.26	6.20	6.14
cutlery_dishes	6.03	5.34	6.05
keychain	4.56	4.53	4.36
coffee_machine	3.78	3.13	3.65
door_open_close	3.57	3.44	3.18
toilet_flushing	3.16	2.82	3.18
wardrobe_drawer_open_close	1.98	1.79	1.68
window_open_close	1.29	1.23	1.16
bell_ringing	1.09	1.40	1.43
light_switch	0.27	0.41	0.26

Table 3: Test macro-F1 leaderboard. Strategy A is the original training set; Strategy B adds audio-domain augmentation (Section 3.5). All numbers use per-class tuned thresholds.

Model	**Strategy A**	**Strategy B**	Δ_{B-A}
Empirical Prior	0.0531	–	–
Logistic Regression	0.4723	0.4749	+0.003
Random Forest	0.5040	0.4982	−0.006
MLP v2	0.5495	0.5534	+0.004
XGBoost	0.5605	0.5564	−0.004
AST (linear probe)	0.5859	–	–
CNN (log-mel)	0.6035	–	–
CRNN (log-mel)	**0.6048**	–	–

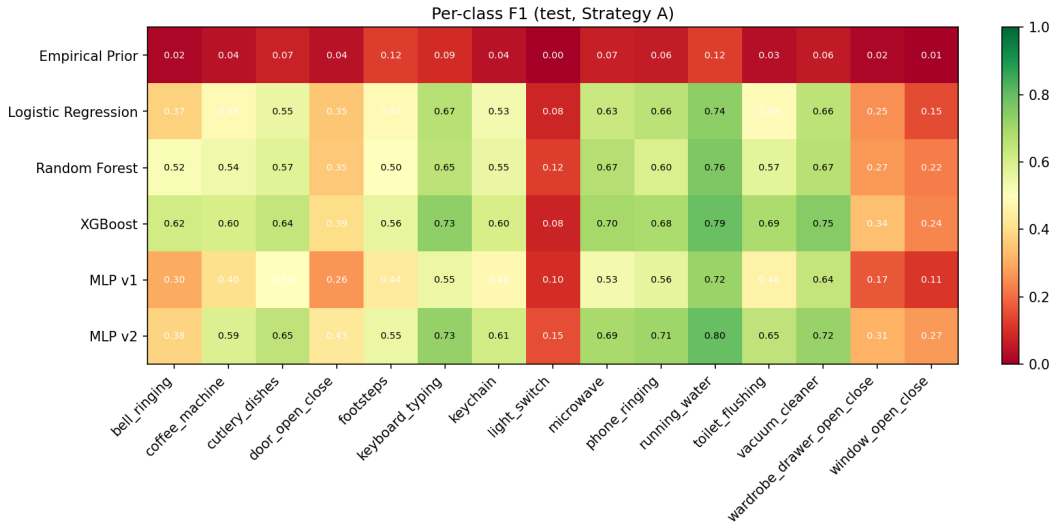


Figure 1: Per-class F1 (test) across the model suite. Sustained broadband classes (`running_water`, `vacuum_cleaner`, `microwave`) are easy at every capacity tier; brief transients (`light_switch`, `window_open_close`, `wardrobe_drawer_open_close`) fail across the board on aggregated features and recover only partially on spectrograms.

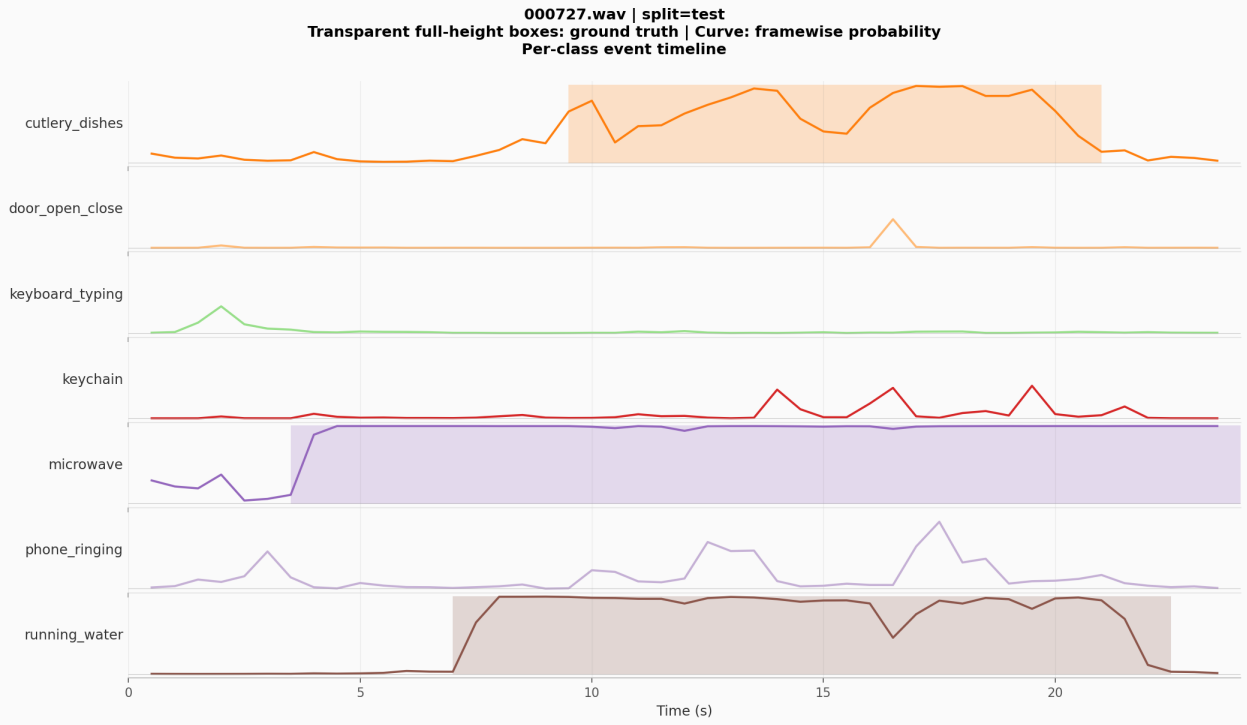


Figure 2: `000727.wav`: three sustained classes detected at F1 0.82–0.99. Transparent boxes are ground truth; coloured curves are CNN per-segment probabilities.

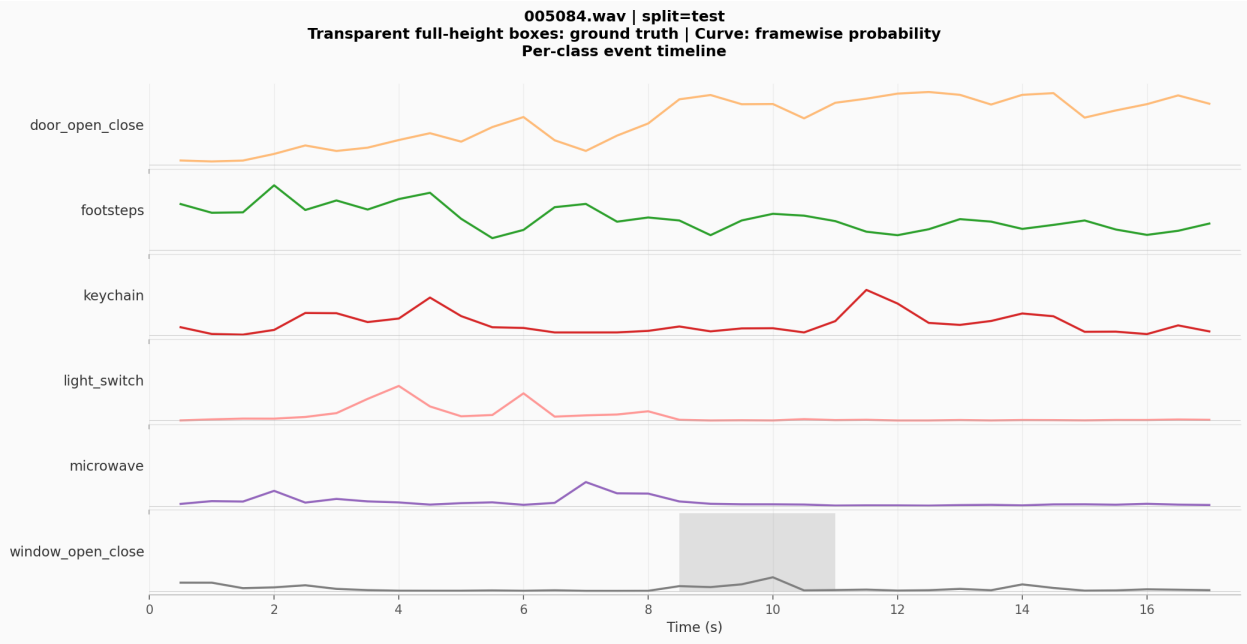


Figure 3: `005084.wav`: the only true class (`window_open_close`, grey box) is missed entirely; the model hallucinates three other brief-transient classes (22 FP segments combined).

6 Disclosure of LLM and AI Tool Use

We used Claude (Anthropic) and Cursor (GPT-4-based) as engineering assistants throughout: generating code scaffolding for the training pipeline, AST extraction, the RF optimisation study, the case-study notebook, and this report draft; iterating on debugging (the MPS `non_blocking` hang in AST extraction, an `np.savez` extension bug, a threshold-transfer bug in the case study); and producing initial drafts of Sections 1 through 6. All numerical results, plots, and conclusions were verified against the raw artefacts on disk. The authors take full responsibility for the methodological choices and the final text.

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